



PHILADELPHIA
— SEPTEMBER 27-30, 2026 —
Rooted in Brotherly Love & Lifelong Community

ANNUAL
MEETING

AACPDM 2026 Pre-Conference

Course PC1 · Session 2

The Promise of Biomarkers

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Epigenetic Biomarkers: Toward Earlier CP Diagnosis

DNA methylation, where we look in the genome, and what blood and muscle can tell us about early brain injury

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AACPDM 2026 Pre-Conference Course PC1 · Session 2: The Promise of Biomarkers



FINANCIAL DISCLOSURE
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1. Disclosure of Relevant Financial Relationships

I have no financial relationships to disclose.

**2. Disclosure of Off-Label and/or investigative
uses:**

I will not discuss off label use and/or investigational
use in my presentation.

The diagnostic gap

Most children are diagnosed well after the window for early intervention opens

~24 mo

This is how long it takes for a process called myelination to be largely completed

3–4 mo

Achievable with GMA + HINE + MRI in high-risk infants at implementing centers

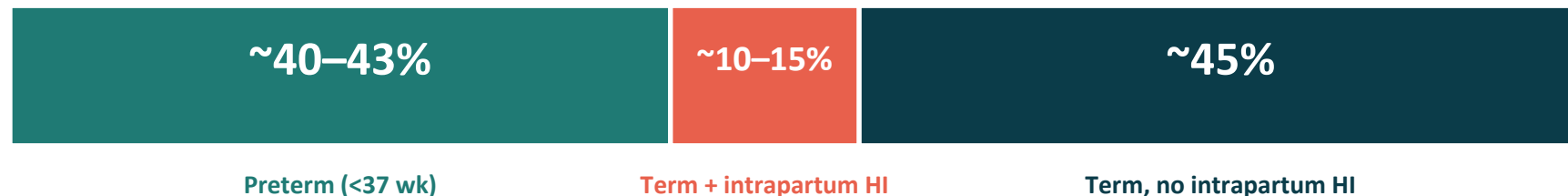
~5 yr

Before diagnostic confidence is high in milder or low-risk presentations

The gaps reflect access and ambiguity: Early-detection tools work best for infants already flagged as high risk and require specialized expertise that many families never reach.

Who develops CP?

Most children with CP were born at term, and few had intrapartum hypoxia–ischemia



Approximate shares of all children with CP in high-income registers; segments combine separate sources and are rounded.

Preterm birth

The strongest single risk factor, yet preterm children are a minority of the CP population. In Australian singletons born 1995–2014, 63% of children with CP were born at term.

Intrapartum HI

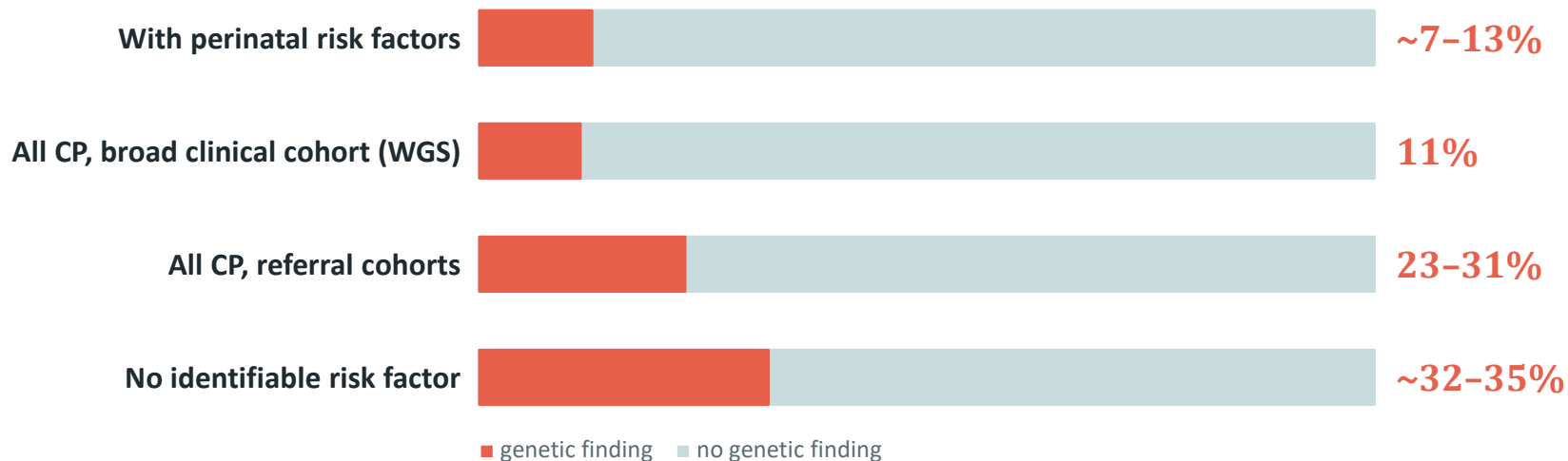
Associated with roughly 10–15% of all CP. Most term-born CP is not associated with intrapartum hypoxia–ischemia.

Everything else

Perinatal stroke, malformations, infection and inflammation, genetic causes, postnatal injury (~5%), and a large unexplained remainder.

Genetics explains some CP — but a minority

Share of children with a reported pathogenic or likely pathogenic finding on exome or genome sequencing



Even without an identifiable perinatal risk factor, about two-thirds of children have no current genetic finding.
And a reported variant is not always the cause of the CP; reported variants may not have predictive value.

Can Epigenetics Fill the Gap?

Layers of regulation that sit above the DNA sequence — and respond to what happens around birth

What is left unexplained?

Most children with CP have no genetic diagnosis, and most were not preterm or exposed to intrapartum hypoxia. Yet something happened.



Epigenetics: Changes how genes are read without altering the DNA sequence itself.

DNA methylation

Methyl groups placed on cytosines, mostly at CpG sites. Copied at cell division, so early patterns persist.

Histone modification

Tags on the proteins DNA wraps around loosen or tighten packing, opening or closing transcription in whole regions.

Non-coding RNA

microRNAs, long non-coding RNAs, and circular RNAs tune which transcripts survive and get translated.

This talk takes up DNA methylation; Dr. Mona Batish returns to RNA regulation in muscle later this morning.

Epigenetics: What is DNA methylation?

A chemical tag on DNA that helps set which genes are on or off



Unmethylated region — gene typically accessible



Methylated region — gene typically silenced

Filled = methyl group present at a CpG site

Does not change the DNA sequence

Acts like annotations in the margin of the genetic text.

Responds to the environment

Hypoxia, infection, and inflammation all leave methylation marks.

Stable and measurable

Can be read from a small blood sample, including dried newborn blood spots.

Why methylation is a plausible CP biomarker

The causes of CP are the same exposures known to reshape the methylome

1

Perinatal insult

Hypoxia-ischemia, infection, inflammation, prematurity

2

Methylation marks

Changes laid down during a highly dynamic developmental window

3

Persisting signature

Early-life patterns stabilize and remain detectable in blood and tissue

Key point: methylation changes rapidly in infancy and then settles. A signal captured early may reflect the original injury; a signal captured years later may also reflect living with CP.

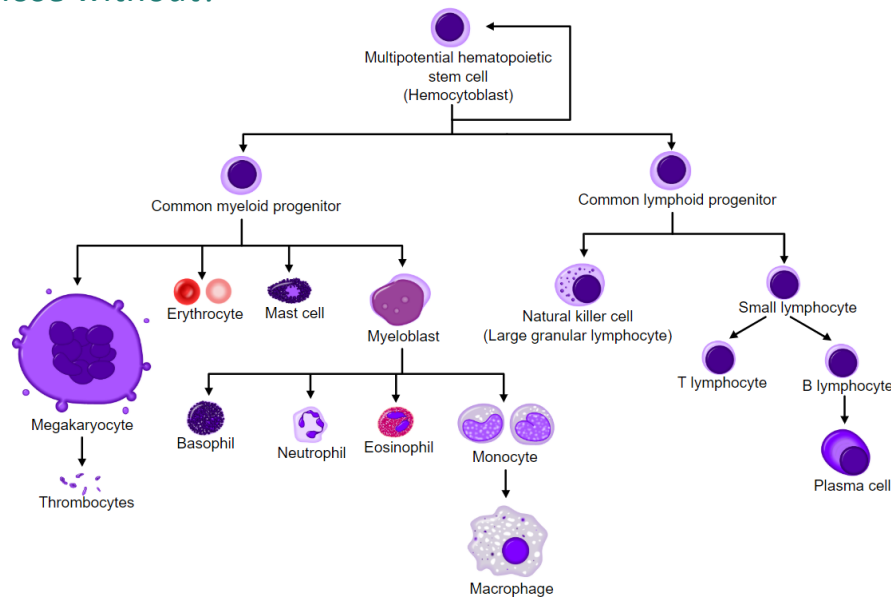
The question for the rest of this talk: Can a mark that records an early insult be turned into a test?

Study Hypothesis for Blood Biomarkers

Can the methylome separate children with CP from those without?

HYPOTHESIS

DNA methylation patterns distinguish a CP cohort from a non-CP cohort.

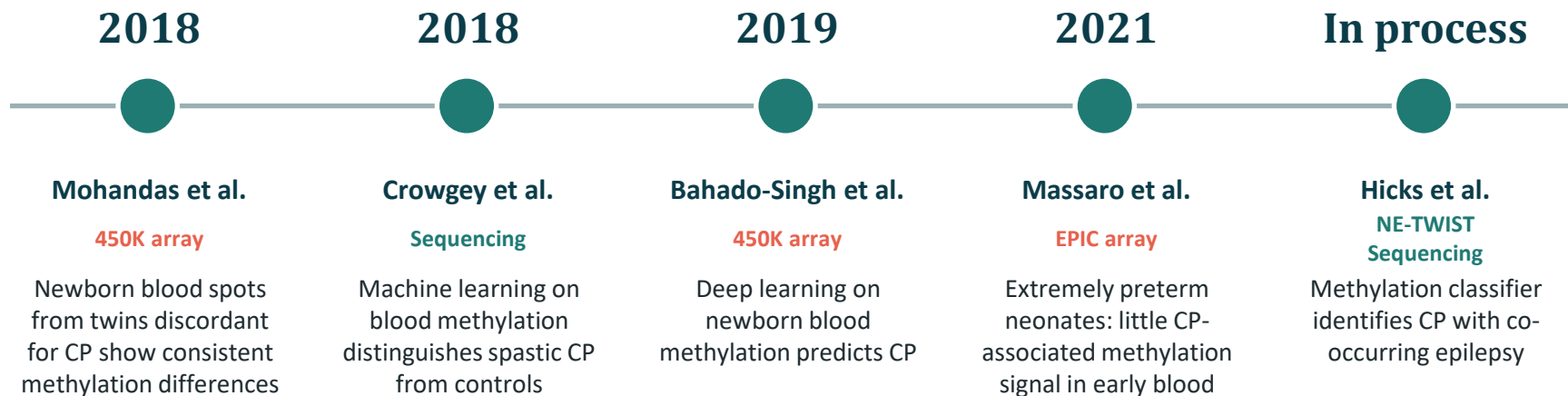


Testable prediction:

A defined set of CpG loci shows reproducible differential methylation between the two cohorts.

The evidence so far

Multiple groups, tissues, and designs point in the same direction



Consistent theme: a signal appears in several studies, but cohorts are small and tissue, age, and **measurement platform** all differ.

Are we looking in the right places?

Most CP methylation studies used arrays that read a small, island-heavy slice of the genome

Illumina 450K  ~450,000 CpGs (~2%)

Illumina EPIC  ~850,000 CpGs (~3%)

Our EM-seq  ~4 million CpGs (~14%)

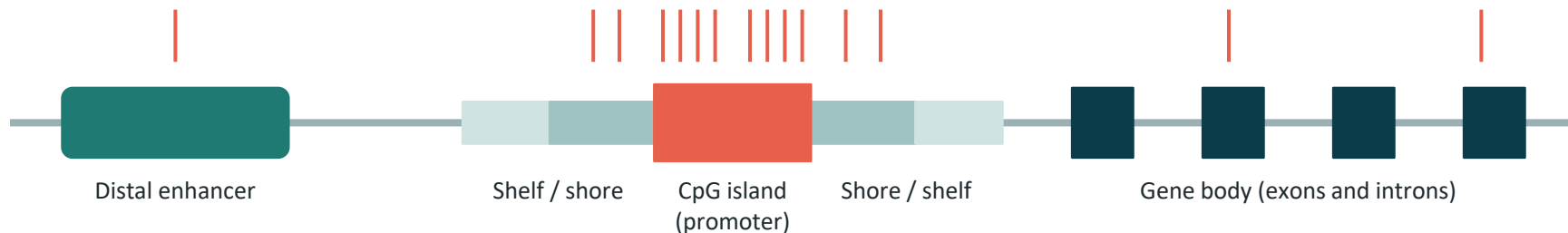
Human genome  ~28 million CpGs

Arrays were built around “CpG islands” and promoters. The 450K array (Mohandas 2018, Bahado-Singh 2019) largely missed distal regulatory elements; EPIC (Massaro 2021) added enhancer probes but still samples only a sliver of them.

Where the informative CpGs may be hiding

Array probes cluster where methylation changes least

Array probes



Where arrays concentrate

CpG islands and promoters: mostly unmethylated in every tissue and relatively stable, leaving limited room to record an early-life insult.

Where signals may live

Shores, distal enhancers, and gene bodies: tissue-specific and environment-responsive. EPIC represents only ~7% of distal regulatory elements, often with a single probe.

Our current blood study

Deep methylation profiling in children with spastic CP and pediatric controls

97

children with spastic CP or
pediatric controls

~4 M

CpG sites by EM-seq, reaching
beyond islands into shores,
enhancers, gene bodies

11

loci in a candidate classifier
panel

Preliminary performance: 80% sensitivity, 90% specificity on a 20-sample holdout.

Now strengthening with cell-composition adjustment, false-discovery control, nested cross-validation, and comparison with simpler models. Expect these numbers to move.

What these data do not yet show

Honest caveats clinicians should hear



Established CP, not infants

Samples come from children already diagnosed. Early detection requires testing infants before diagnosis.



Cause vs. consequence

Years of altered mobility, medication, and surgery can shape methylation too.



Confounders

Blood cell mix, anticonvulsants, prematurity, and age all move methylation.

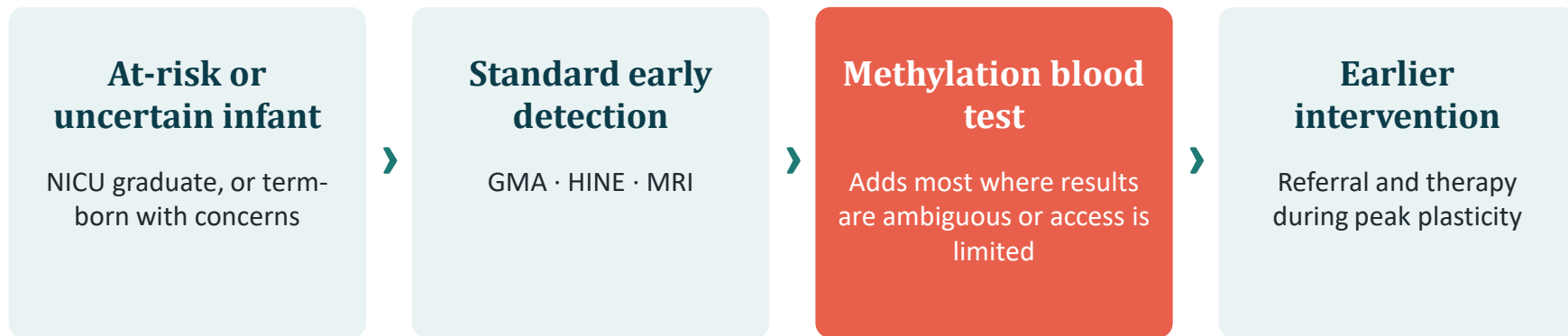


Small cohorts

Every study to date needs independent, multi-site replication.

Where a methylation test could fit

A complement to GMA, HINE, and MRI — not a replacement



Why a complementary test matters: the same test performs very differently depending on how likely a CP diagnosis is before testing. A blood test is most useful for intermediate-risk infants, where MRI and exam leave real uncertainty — and for families far from specialized centers.

How muscle tells the story of brain injury

CP muscle is not just a bystander to the brain lesion



Muscle stem cells are different

Satellite-cell–derived progenitors from children with spastic CP carry distinct methylation patterns.

Robinson et al., J Pers Med 2022



Gene programs are altered

Muscle and satellite cells show shifted transcriptional profiles in spastic CP.

Robinson et al., DMCN 2021



Regulatory RNAs are involved

circNFIX regulates MEF2C, a key muscle-development factor, in CP satellite cells.

Romero et al., J Biol Chem 2024

Working model

Perinatal inflammation / HI



Mitochondrial stress in muscle



Succinate & fumarate rise



TET demethylases inhibited



Lasting methylation changes in muscle gene programs



Take-home messages

- 1 DNA methylation records early-life exposures, including the insults that cause CP.
- 2 Where we look matters: arrays read ~2–3% of CpGs, mostly islands; enhancers and gene bodies may hold the signal.
- 3 Early classifiers are promising, but not ready for clinical use.
- 4 The critical next step is prospective testing in infants before diagnosis, and clinicians can help.

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